1. Clinical Question & Core Hypothesis

Clinical Question

Intravesical BCG provokes local immune infiltration (TILs) and upregulates PD-L1, yet many urothelial carcinomas fail to respond; why does TIL presence sometimes not translate into BCG clinical benefit? 2) Tier 1 paths used: TMPRSS2/ERG → cooperativity with PI3K pathway activation → increased PD-L1 (CD274). I built on this because it provides a direct tumor-intrinsic route (oncogenic fusion → PI3K signaling) to immune-checkpoint upregulation that can negate local immune effects. 3) Tier 2 evidence supporting the claim: BCG itself up-regulates PD-L1 and binds urothelial cells (providing the setting where tumor PD-L1 matters); PSMA/FOLH1 and prostate-stem-cell links indicate how tumor lineage/oncogenic context can shape immune-modulatory programs; androgen-axis/AR signaling and PTEN/PIK3CA alterations are plausible modifiers of PI3K–PD-L1 coupling. 4) Why therapy_response endpoint: the mechanistic chain centers on BCG-induced immunity and PD-L1–mediated suppression, so 'therapy response to BCG' is the most direct, actionable clinical endpoint and has immediate translational implications (predicting and reversing BCG failure). 5) Deliberate exclusions: I excluded high-level nodes (Carcinoma, bladder organ anatomy) and peripheral prostate-treatment nodes (androgen deprivation) from the core hypothesis to keep focus on the tumor-intrinsic oncogenic → immune-checkpoint axis; AR and PSMA/lineage signals are retained only as secondary modifiers rather than core anchors?

Working Hypothesis

In urothelial carcinoma treated with intravesical BCG, high TIL density alone does not guarantee an anti-tumor response if tumor-intrinsic PI3K pathway activation is present, as indicated by PTEN loss and/or PIK3CA mutation leading to PI3K-AKT-mTOR–driven upregulation of PD-L1 (CD274) on tumor cells. This PD-L1 overexpression suppresses infiltrating T cells, producing a dysfunctional immune milieu that undermines BCG efficacy. Therefore, tumors that are PD-L1 high (CD274 high) and PI3K-activated (PTEN loss and/or PIK3CA mutation) with prominent TILs will show poorer response to intravesical BCG compared with tumors lacking PI3K activation. This mechanistic axis predicts that combining BCG with PD-1/PD-L1 blockade and/or PI3K pathway inhibition will restore or enhance BCG responsiveness in TIL-high, CD274-high, PI3K-activated urothelial tumors.

Element	Value
Imaging feature	Tumor-Infiltrating Lymphocyte (TIL) density
Anchor gene(s)	TMPRSS2-ERG, CD274
Primary endpoint	therapy_response

2. Why This Gene — Not Others

Primary hypothesis gene(s)

TMPRSS2-ERG: Selected as a core mechanistic driver from Tier 1 knowledge graph analysis.

CD274: CD274).

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core

Secondary	PIK3CA	Supporting modifier from knowledge graph
Secondary	PTEN	Supporting modifier from knowledge graph
Secondary	FOLH1	Supporting modifier from knowledge graph
Secondary	AR	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: In retrospective cohort data, the prognostic benefit of the imaging feature will be modified by the anchor gene expression level (significant interaction term).

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Interaction-first biomarker strategy — asks WHY the imaging feature sometimes fails, not just whether it predicts outcome.

Caveats & Risks

- Gene expression does not equal protein activity; phospho-protein or functional assays are needed for confirmation.

Alternative Directions

- PIK3CA: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- PTEN: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.